William To

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Professional Summary

Senior Software Engineer with 4+ years building secure, cloud-native distributed systems in regulated, high-traffic environments. Specialises in scalable .NET microservices, high-performance REST and GraphQL APIs, and complex data integrations across AWS and Azure. Shapes service architecture across squads, mentors junior engineers, and champions AI-assisted engineering practices that measurably improve delivery speed and code quality.

Skills

Languages & Frameworks: C#, .NET Core, ASP.NET Core, Python, TypeScript, React, Next.js

APIs & Architecture: REST, GraphQL, Microservices, Clean Architecture, CQRS, EF Core, Autofac, FluentValidation

Cloud & Infrastructure: AWS (ECS, Lambda), Azure (Functions, DevOps), Docker, Terraform, Bicep, CI/CD, Linux, IIS

Security & Observability: OAuth2, JWT, RBAC, Secure API Design, Automated Testing, Performance Profiling

Frontend: React, Next.js, Tailwind CSS, Zustand, Apollo Client, PWA

AI-Assisted Engineering: Cursor, Claude Code, AI-accelerated refactoring, test generation, documentation

Work Experience

PicNet, Sydney Jan 2024 – Jul 2025

Senior Software Engineer / Backend Technical Lead

- Served as primary backend SME across multiple delivery squads, leading design and delivery of scalable .NET and Python microservices, secure REST/GraphQL APIs, and downstream integrations supporting high-throughput production workloads.
- Redesigned multi-terabyte SQL schemas, indexes, and execution plans for NSW Health, reducing critical reporting pipelines from hours to minutes and achieving near-zero downtime during migration.
- Led migration of legacy monolithic platforms to serverless and containerised architectures using Azure Functions, AWS ECS, Docker, and Terraform, cutting deployment and onboarding effort by **70%**.
- Engineered real-time financial data synchronisation pipelines between Lightspeed, Xero, and internal systems, ensuring transactional consistency across billing, payments, and downstream reporting.
- Defined service boundaries, API contracts, and integration patterns across squads; aligned architecture decisions with product owners and engineering leads to reduce cross-team dependencies and delivery risk.
- Drove adoption of AI-assisted development tools (Cursor, Claude Code) for refactoring, test generation, and documentation, improving team throughput while maintaining enterprise security and quality standards.

PicNet, Sydney Feb 2022 – Jan 2024

Software Engineer *(Promoted from Graduate)*

- Rebuilt high-volume patient-data ingestion services with row-level validation and RBAC enforcement, improving throughput **4×** and reducing ingestion failures to near-zero across 600+ concurrent users.
- Built metadata-driven Azure Functions automating SharePoint-to-SQL data ingestion for Guide Dogs Australia, reducing reporting cycle time by **80%** and eliminating manual data handoffs.
- Delivered React-based analytical dashboards and Power BI integrations, increasing analyst engagement and self-service reporting adoption by **60%**.

OpenHomeLoan Aug 2020 – Apr 2021

Software Developer Intern

- Built ASP.NET automation services integrating SalesTrekker APIs into the loan processing pipeline, tripling approval throughput and enabling scalable personalised lending workflows at volume.
- Developed and shipped responsive marketing pages that increased campaign conversion rates by **25%**.

Education

University of Technology Sydney (UTS)
Bachelor of Engineering (Honours) — ICT/Software & Diploma in Professional Engineering Practice ***GPA: 6.2 / 7.0***

Project Work

FitNote — Full-Stack Fitness Tracking Platform fitnote-webapp.vercel.app

.NET, ASP.NET Core, PostgreSQL, GraphQL, Next.js, React, Tailwind, Apollo Client, Zustand, PWA

- Designed and shipped a production-grade fitness platform with offline-first PWA support, real-time workout tracking, and a searchable 400+ exercise library used by active users.
- Built a scalable GraphQL API with schema-driven validation, CQRS-style resolvers, and field-level JWT authorisation, applying the same security patterns used in regulated enterprise environments.
- Architected the backend using Clean Architecture principles with EF Core, Autofac dependency injection, and Fluent-Validation, maximising testability and long-term maintainability.
- Engineered a React frontend with predictable Zustand state management, accessible UX patterns, and responsive layout across device breakpoints.

Test Platform — Multi-Tenant Assessment System github.com/willyyto/TestSystem

.NET, React, SQL Server, JWT, RBAC, CI/CD

- Built a multi-tenant SaaS assessment platform supporting tenant isolation, role-based access control, timed assessments, and secure JWT authentication.
- Designed backend services for predictable multi-tenant query performance and implemented automated CI/CD pipelines enabling safe, repeatable production deployments.

References

Available upon request.